**QP CODE: 24900104**

**MAHATMA GANDHI UNIVERSITY KOTTAYAM**
**FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION**
**NOVEMBER 2024**
**MG1DSCFNN100- BASIC NUTRITION FOR HOTEL OPERATIONS**

**PART - A**
**Answer any 10 questions, each questions carries 2 marks**

1. Carbohydrates, Protein, Fat, Vitamins and Minerals
2. Excess, deficient or imbalance of nutrients
3. It is a complete state of physical, mental, social wellbeing and not merely the absence of disease or infirmity.
4. Nutritional status refers to the evaluation of an individual's health in terms of their diet, body weight, and biochemical data.
5. 1. Complete protein- it contains all the essential amino acids, 2. Incomplete proteins-Lack one or more essential amino acids, 3. Partially incomplete protein- Contain essential amino acids but not enough to meet metabolic needs
6. Helps the immune system. Wound healing, iron absorption, formation of connective tissue, act as antioxidant, prevents scurvy
7. Tertiary structure of protein- The arrangement and interconnection of proteins into specific loops and bends forms the tertiary structures. This structure contains hydrogen, ionic and disulfide bonds.
8. A non essential amino acid is an amino acid that can be synthesized within the body
9. Energy providing, bodybuilding, and protection and regulation
10. Hollow calorie foods, also known as empty calorie foods, are high in calories but low in nutrients
11. Plan for several smaller meals throughout the day or larger meals at specific times.
12. Factors affecting meal planning- dietary restrictions and preferences, nutritional requirements, time constraints, budget, seasonal availability, cooking skills and equipment, cultural and regional influences, meal frequency and portion control, time of day, personal preferences and creativity.
13. Carbohydrates, Protein , Fat, Vitamins, Minerals, Fibers and water
14. Creating a budget and shopping for ingredients on sale can help reduce costs
15. Lipids- 1. Animal origin, 2. Marine origin, 3. Vegetable origin

**PART -B**
**Answer any 4 questions, each questions carries 5 marks**
16. Functions of food- 1. Physiological function- Energy providing, Bodybuilding, Protection and regulation, 2. Psychological functions- sense of comfort, security, love and attention, token of friendship and acceptance, used for reward and punishment, 3. social functions- Food serves as a reason for getting together, Temples, gurdwara serve same food to all, Birthday, marriage and others celebrations become reasons to serve food.
17. Functions of protein- maintains proper ph., balances fluids, strengthens immune health, transports and stores nutrients, provides energy.

18. Vitamin B complexes- 1. Vitamin B1(thiamine)- Essential for release of energy from carbohydrates, needed for normal functioning of nervous system, sources- milk, meat , fortified cereals, egg. 2. vitamin B29 (riboflavin)- metabolism of carbohydrates, growth, repair, development of body tissues. Sources- offal, milk, egg, cheese. 3. Vitamin 3(Niacin)- metabolism of carbohydrates, protein and fat, normal functioning of nervous tissues. Sources-meat, offal, bran, wheat, dry fruits. 4.Vitamin B5( pantothenic acid)- Helps metabolize carbohydrates, proteins, fats, and alcohol, and is important for brain and nervous system health. 5.Vitamin B9- RBC formation, Reduces the neural tube defects, development of brain spinal cord, skeleton in foetus

19. 1. Psychological functions- sense of comfort, security, love and attention, token of friendship and acceptance, used for reward and punishment, 2. social functions- Food serves as a reason for getting together, Temples, gurdwara serve same food to all, Birthday, marriage and others celebrations become reasons to serve food.

20. Basal Metabolic Rate (BMR) is the amount of energy required by the body to maintain basic life-sustaining functions( involuntary activities), such as breathing, blood circulation, and organ function. Factors: 1. Age, 2. gender, 3. Body composition, 4. Body size, 5. Hormones, 6. Diet.

21. Vitamin D- It is a fat-soluble nutrient that helps the body absorb calcium, magnesium, and phosphate, to build strong bones and teeth, and helps prevent osteoporosis,helps muscles move and nerves carry messages, helps the immune system fight off bacteria and viruses, helps to prevent rickets

**PART- C**

**Answer any 1 question, each question carries 10 mark**

22. Carbohydrates- Carbohydrates are the most abundant organic molecules in nature and the major functional constituents of living cells. A carbohydrate is a biomolecule consisting of carbon (C), hydrogen (H) and oxygen (O) atoms, usually with a hydrogen–oxygen atom ratio of 2:1 (as in water). Classification- Monosaccharides-The word "Monosaccharides" derived from the Greek word "Mono" means Single and "saccharide" means sugar. Monosaccharides are simple sugars. They are sweet in taste. They are soluble in water. They are crystalline in nature. Eg; glucose, fructose, galactose 2. Disaccharides-Disaccharides are produced from the condensation of 2 monosaccharide molecules. Eg; lactose, maltose, sucrose. 3. Oligosaccharides- Oligosaccharides consist of short chains of 3-9 monosaccharide units 4.Polysaccharides- Polysaccharides contain hundreds or thousands of carbohydrate units. (Homopolysaccharide and Heteropolysacchrides)

**Functions**: Provides energy, protein sparing action, normal fat metabolism, provide fibre, regulate blood glucose and provide carbon skeleton for synthesis of aminoacids. Sources; milk, cereals and millets, legumes, starchy vegetables, beverges.

23. BMR- Basal Metabolic Rate (BMR) is the amount of energy required by the body to maintain basic life-sustaining functions( involuntary activities), such as breathing, blood circulation, and organ function. Energy expenditure-Energy expenditure is the amount of energy the body uses for physiological processes, physical activity, and changes in body composition. **Factors**: 1. Age, 2. gender, 3. Body composition, 4. Body size, 5. Hormones, 6. Diet.

24. Meal planning is defined as a simple process which involves the application of the knowledge of food, nutrient, food habits, likes and dislikes to plan wholesome and attractive meals

**Importance of meal planning**- Meet the nutritional needs of the individuals who will be consuming the food, Plan meal within the food budget, Simplify purchase, preparation, and storage of foods, Provide attractive, appetizing meal with no

monotony, Save time and money, Minimize the overhead expenditure, i.e fuel, electricity, water and labour, Reduce mealtime stress
Principles of meal planning: Adequacy, balance, calorie control, nutrient density, variety, moderation.